Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

Kamiar Asgari and Michael J. Neely

***Abstract*—We study online control of a scalar storage system with nonnegative adversarial resource arrivals and state-dependent action constraints. The cost is adversarial, time-varying, and depends on both the action and the state. At each time, the controller selects a feasible action before the current resource arrival and cost are revealed. To handle the resulting coupling between feasibility and online learning, we introduce *Simplex Disturbance-Action Control* (SDAC), an H-dimensional finite-memory policy class parameterized by a subprobability simplex. The key feature of SDAC is that every fixed policy in the class is feasible by construction. Our adaptive controller updates these parameters by entropic online mirror descent while preserving feasibility along the time-varying trajectory. We prove regret $O\left(\sqrt{T\log(H+1)}\right)$ relative to the best fixed SDAC policy. As the retention coefficient α approaches one, the leading factor scales as $\Theta((1-\alpha)^{-3/2})$. We also derive a minimax lower bound which, when T is at least on the order of $(1-\alpha)^{-1}$, scales as $\Omega((1-\alpha)^{-1}\sqrt{T})$ near $\alpha = 1$, leaving a factor $(1-\alpha)^{-1/2}$ between the upper and lower bounds. We also show that SDAC geometrically approximates a decreasing-allocation policy class that spreads each arrival over time according to a common decreasing allocation sequence, a structure that arises in stochastic energy-harvesting control. Geometric allocation sequences induce all feasible proportional state-feedback policies. With $H = \Theta(\log T)$, the resulting regret against the decreasing-allocation class, and hence against feasible proportional state-feedback policies, is $O\left(\sqrt{T\log\log T}\right)$. The framework applies to energy-harvesting batteries and other storage-constrained resource systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY resource-allocation systems have a storage state that changes over time and limits the control action at each time. An action that lowers the current cost may leave less resource for future decisions, while conserving the resource may increase the current cost. The controller must balance these effects while satisfying the action constraint at every time.

We study this problem for a scalar storage system that loses a fixed fraction of its stored resource each period. At each time, the controller observes the current state and all past arrivals and cost functions, chooses a control action, and then observes the current arrival and cost function. The arrival and cost sequences may be chosen adversarially based on past observations; we do not assume a known probability model. Our goal is to construct a feasible online controller whose cumulative cost is close to that of the best fixed policy, chosen in hindsight, from a specified benchmark class.

The main difficulty is that the current action affects both the next state and the actions that will be feasible later. Therefore, the controller cannot update its decisions independently at each time. *Disturbance-Action Control* (DAC) policies address this dependence by combining a stabilizing state-feedback term with a linear function of past disturbances. In our storage model, however, unrestricted filter weights can produce a negative action or an action larger than the available resource.

We address this issue by introducing *Simplex Disturbance-Action Control* (SDAC). SDAC restricts the filter weights to a subprobability simplex and includes powers of the retention coefficient in the filter. These choices ensure that every fixed SDAC policy satisfies the action constraint. The induced state and action are also affine in the policy parameter, so a convex state–action cost is convex as a function of that parameter. The parameter can therefore be updated using online convex optimization.

B. Model and Information Pattern

The storage state $x_t \geq 0$ evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t,$$

where $\alpha \in (0,1)$ is a fixed retention coefficient, u_t is the control action, and $w_t \in [0,1]$ is a nonnegative resource arrival. The action must satisfy

$$0 \leq u_t \leq \alpha x_t.$$

Thus, the controller cannot use more than the amount that remains after the storage loss.

At time t, the controller chooses u_t using the current state and the information available through time $t-1$. The arrival w_t and the convex cost function c_t are revealed after the action is selected, and the controller incurs the cost $c_t(x_t, u_t)$. The pair (w_t, c_t) may depend on the observations through time $t-1$, but not on the current action. Both the online trajectory and every benchmark trajectory must satisfy the same dynamics and action constraint. The formal model and assumptions are given in Section III.

Energy-harvesting battery management is our main application. In this case, x_t is the battery state of charge, w_t is the harvested energy, u_t is the energy used, and $1-\alpha$ is the fraction of stored energy lost during one period. Similar dynamics arise in perishable inventory systems, water reservoirs with

K. Asgari and M. J. Neely are with the Department of Electrical and Computer Engineering, University of Southern California, Los Angeles, CA 90089 USA (e-mail: kamiaras@usc.edu; mjneely@usc.edu).

Corresponding author: K. Asgari.

storage loss, and queues with abandonment. In many standard queueing models, $\alpha = 1$; that case is outside our main results and is discussed in Section VIII.

C. Contributions

Our contributions are as follows:

- **Policy class.** We introduce SDAC, a finite-memory policy class parameterized by a subprobability simplex. We prove that every fixed SDAC policy is feasible for every arrival sequence with entries in $[0, 1]$. We also show that the state and action generated by a fixed SDAC policy are affine in its parameter. Under our convexity and Lipschitz assumptions on the state–action costs, the induced cost is convex and Lipschitz as a function of the policy parameter.
  We further show that SDAC approximates the decreasing-allocation policy class defined in Section IV-C. This class allocates each arrival over future periods according to the same decreasing sequence. It contains every feasible proportional state-feedback policy $u_t = kx_t$, with $k \in [0, \alpha]$, and the approximation error decreases geometrically with the memory horizon H.
- **Online control and regret.** We develop an online controller that updates its SDAC parameter using entropic online mirror descent (OMD) over the subprobability simplex. Because the parameter changes over time, the nominal SDAC action is clipped when needed to satisfy the current action constraint. We prove that this controller remains feasible and achieves
  $$O\Big(\sqrt{T \log(H + 1)}\Big)$$
  regret against the best fixed SDAC policy; see Theorem 1. For fixed L_x and L_u, the leading factor scales as $\Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$.
  For fixed α, take the memory horizon H to grow as $\log T$. The resulting regret is $O(\sqrt{T \log\log T})$ against the decreasing-allocation class. The same bound holds against the best feasible proportional state-feedback policy; see Corollaries 2 and 3.
- **Minimax lower bound.** We prove a lower bound of order $\sqrt{T}$ on the expected regret of every causal feasible controller, including randomized controllers, even when the arrival and cost sequences are deterministic and fixed in advance; see Theorem 2. When T is at least on the order of $(1 - \alpha)^{-1}$, the lower bound scales as $\Omega((1 - \alpha)^{-1}\sqrt{T})$ near $\alpha = 1$; see Corollary 4 and Remark 1. The dependence of the upper and lower bounds on $1-\alpha$ therefore differs by a factor $(1-\alpha)^{-1/2}$. Closing this gap is left for future work.

D. Related Work

a) Online nonstochastic control: Online nonstochastic control studies dynamical systems with adversarial disturbances and costs. Performance is measured by regret against a policy chosen in hindsight, under suitable stability conditions [1], [2]; see [3] for an overview. For known linear systems, Simchowitz et al. establish the optimal $\sqrt{T}$ dependence on the horizon while leaving some dependence on the system's stability unresolved [4]. Related work also considers partial observability [5] and bandit feedback [6].

A central idea in this literature is DAC. A DAC policy combines a stabilizing state-feedback action with a finite linear filter of past disturbances. The action is therefore described by finitely many policy parameters that can be updated using online methods. SDAC uses the same basic idea but restricts the parameters in a way suited to storage systems with nonnegative arrivals.

Online control with affine or more general constraints on states and actions has been studied in [7], [8]. The constraint $u_t \le \alpha x_t$ is affine in the state and action, but the range of allowed actions depends on the current storage state. Our approach uses the structure of the storage dynamics to ensure that every fixed SDAC policy is feasible and that the online controller remains feasible while its parameter changes.

Our analysis also considers how regret depends on the retention coefficient α. Kumar et al. study online convex optimization in which past decisions can affect all later costs, and derive bounds that depend on how quickly those effects decay [9]. In their model with geometrically decaying memory, the lower bound scales as $\sqrt{T}$ divided by one minus the decay factor, paralleling our $\Omega(\sqrt{T}/(1-\alpha))$ lower bound for sufficiently long horizons; see Remark 1. Our lower bound is proved directly for state–action costs under the state-dependent action constraint.

b) Energy harvesting: Several energy-harvesting policies adjust energy use according to the available battery level. Shaviv and Özgür [10] study a policy that uses a fixed fraction of the available energy and prove guarantees under independent and identically distributed (i.i.d.) energy arrivals. Arafa et al. [11] consider more general sequences for spending energy after a recharge. Zibaeenejad and Chen [12] derive a related decreasing spending rule when future arrivals are known in advance.

These works motivate the *decreasing-allocation policy class* defined in this paper. Under such a policy, each arrival is allocated over future periods according to the same decreasing sequence. In our model, this allocation begins one period after the arrival because the controller does not observe the current arrival before choosing the action.

The models in [10]–[12] use random arrivals from a fixed model, reveal the current harvested energy before allocation, and optimize a fixed reward that depends on the allocated power. In our setting, the arrival and cost sequences may be chosen adversarially, the controller acts before observing the current arrival and cost, the cost can depend on both the storage state and the action, and performance is measured by regret against a fixed policy.

Yu and Neely [13] study expected utility maximization under i.i.d. system states. Asgari and Neely [14] obtain an $O(\sqrt{T})$ regret–battery-capacity tradeoff for arbitrary loss sequences while retaining i.i.d. energy arrivals. Both works use objectives that depend only on the action and compare against a fixed action. Here, the arrivals may be chosen adversarially, the cost depends on the state and action, and the comparator is a fixed policy rather than a fixed action.

Lin et al. [15] study online age-of-information optimization in an energy-harvesting system with related battery dynamics. They compare their controller with an unrestricted offline optimum using a competitive ratio. Their objective depends on the energy allocation but not directly on the battery state, and the current arrival is observed before the allocation. Our controller acts before observing the current arrival, and our benchmark is a fixed policy chosen in hindsight.

c) Queueing and network optimization: Queueing control also involves states that change through arrivals and service decisions. Classical work studies throughput and queue stability [16]–[18], while stochastic network optimization studies tradeoffs between time-average objectives and queue backlog [19]. Our setting instead allows adversarial arrival sequences and convex state–action costs, and measures regret against a policy chosen in hindsight.

In many queueing models, $\alpha = 1$ because stored packets do not disappear over time. Such models are not direct examples of the $\alpha \in (0, 1)$ system studied here. A queue with abandonment or expiration can have $\alpha < 1$ and is more closely related to our model. Section VIII discusses the case when $\alpha \geq 1$.

Khojastepour and Sabharwal [20] study a fixed scheduler that can be written as a linear filter of past arrivals. Their scheduler minimizes average transmit power subject to a maximum-delay guarantee. In contrast, our controller updates its filter weights online in response to convex state–action costs revealed after each decision.

d) Inventory and reservoir management: Yang et al. [21] study online procurement under inventory constraints. Their controller observes the current demand and price and then chooses how much supply to purchase. Performance is measured by a competitive ratio for the total procurement cost. Our controller instead chooses how much stored resource to use before observing the current arrival and cost.

Hihat and Fermanian [22] consider inventory systems with more general state-transition functions and study policies that replenish inventory up to a target level. Their model is broader in its transition dynamics, but it has a different decision order and a different policy class. Our scalar linear dynamics allow explicit feasibility, approximation, and regret guarantees.

Classical reservoir models also use storage dynamics with random inflows and controlled releases [23], [24]. That literature mainly studies long-run performance under stochastic inflows. It provides another example of control under storage constraints, although physical reservoir models may also include capacity and overflow effects that are not part of our main model.

E. Paper Organization

Section III states the storage dynamics, information pattern, cost assumptions, feasible domain, and regret benchmark. Section IV introduces the SDAC policy class and proves its feasibility and convexity properties. It also shows how SDAC approximates decreasing-allocation policies and proportional state-feedback policies. Section V presents the feasible online controller based on action clipping and entropic OMD. Section VI proves the regret upper bound and extends it to the larger policy classes. Section VII gives the minimax lower bound. Section VIII discusses extensions to $\alpha \geq 1$, vector actions, and multiple storage states. Section IX concludes the paper. Appendices A–D contain the deferred proofs, grouped by the section in which the corresponding results are stated or used.

II. Notation

For every positive integer n, we write $[n] \triangleq \{1, \ldots, n\}$ and $[z]_+ \triangleq \max\{z, 0\}$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is denoted by v_i. If the vector already carries a subscript, the coordinate is appended after a comma: the i-th coordinate of $\mathbf{v}_t$ is $v_{t,i}$. We write $\mathbf{1}$ for the all-ones vector, with its dimension clear from context, and $\langle \cdot, \cdot \rangle$ for the Euclidean inner product. We use $\log$ for the natural logarithm and set $w_\tau \equiv 0$ for $\tau \leq 0$.

A fixed SDAC policy with parameter $\mathbf{m}$ is denoted by $\pi_{\mathbf{m}}$, with induced state and action trajectories $x_t(\mathbf{m})$ and $u_t(\mathbf{m})$. A feasible proportional state-feedback policy with gain $k \in [0, \alpha]$ induces $x_t(k)$ and $u_t(k)$, while a decreasing allocation sequence $\mathbf{q}$ induces $x_t(\mathbf{q})$ and $u_t(\mathbf{q})$. After the online algorithm is introduced, x_t and u_t without arguments denote the realized online trajectory.

III. Model and Problem Setup

Fix a positive integer T and a known retention coefficient $\alpha \in (0, 1)$. The state $x_t \in \mathbb{R}_{\geq 0}$ is the resource level at the beginning of time t. The system starts empty:

$$x_1 = 0. \tag{1}$$

At each time $t \in [T]$, the controller chooses an action $u_t \in \mathbb{R}_{\geq 0}$ satisfying

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

Here, $1 - \alpha$ is the fraction of the stored resource lost during one period. The constraint (2) ensures that the action does not exceed the amount remaining after this loss.

Although w_t is commonly called a disturbance in the online-control literature, we call it an *arrival* because it is nonnegative. We retain the standard names Disturbance-Action Control and Simplex Disturbance-Action Control. The nonnegativity of w_t is important both for the storage interpretation and for the feasibility properties of SDAC.

At time t, the pair (w_t, c_t) may depend on the observations through time $t-1$, but not on the current action or any random choice made by the controller at time t. The controller chooses u_t before observing w_t or c_t. After the action is selected, the arrival and the entire cost function are revealed, the controller incurs $c_t(x_t, u_t)$, and the state is updated according to (3).

For every feasible trajectory,

$$0 \leq x_{t+1} \leq \alpha x_t + 1.$$

Since $x_1 = 0$, induction gives

$$0 \le x_t \le \frac{1-\alpha^{t-1}}{1-\alpha} \le \frac{1}{1-\alpha}, \qquad t \in [T+1].$$

Thus, the storage state is uniformly bounded. In particular, a storage device with capacity $1/(1-\alpha)$ is sufficient to realize every feasible trajectory without overflow. The state and action constraints imply that every feasible state–action pair belongs to the compact set

$$\mathcal{D}_\alpha \triangleq \left\{(x,u) : 0 \le x \le \frac{1}{1-\alpha}, 0 \le u \le \alpha x\right\}. \tag{4}$$

Each revealed cost function has the form

$$c_t : \mathbb{R}^2_{\ge 0} \to \mathbb{R}$$

and is convex in (x,u). We assume that it satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there are constants $L_x, L_u > 0$, independent of t, such that

$$|c_t(x,u) - c_t(x',u')| \le L_x|x-x'| + L_u|u-u'| \tag{5}$$

for all $(x,u),(x',u') \in \mathcal{D}_\alpha$. We also assume access, at every $(x,u) \in \mathcal{D}_\alpha$, to a cost subgradient

$$\boldsymbol{\zeta}_t(x,u) = \big(\zeta_t^x(x,u), \zeta_t^u(x,u)\big) \in \partial c_t(x,u)$$

satisfying

$$|\zeta_t^x(x,u)| \le L_x, \qquad |\zeta_t^u(x,u)| \le L_u.$$

These subgradient bounds hold, for example, when (5) extends to an open neighborhood of $\mathcal{D}_\alpha$; see [25, Thm. 9.13].

Let $\{(x_t,u_t)\}_{t=1}^T$ denote the trajectory generated by the online controller. Its cumulative cost is

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^T c_t(x_t,u_t). \tag{6}$$

Fix a positive integer H, and let $\mathbf{m}$ range over the SDAC parameter set $\mathcal{M}_H$, defined in Section IV. For each $\mathbf{m} \in \mathcal{M}_H$, let $x_t(\mathbf{m})$ and $u_t(\mathbf{m})$ denote the trajectory generated by the fixed policy $\pi_{\mathbf{m}}$ under the same arrival sequence. Define its cumulative cost by

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^T c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big).$$

Both the online trajectory and every benchmark trajectory are required to satisfy (2) under (3). The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_H} J_T(\mathbf{m}). \tag{7}$$

The online controller may update its SDAC parameter over time, while the benchmark in (7) is the best single fixed SDAC policy chosen in hindsight. Here, fixed means that $\mathbf{m}$ is held constant; the policy's actions still respond to past arrivals. Our objective is to satisfy (2) at every time and achieve sublinear regret.

IV. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

DAC combines a stabilizing state-feedback term with a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate losses and efficient online updates. In our scalar storage model, specializing the baseline feedback term to zero gives the representative form

$$u_t = \sum_{i=1}^H \theta_i w_{t-i}, \tag{8}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1,\ldots,\theta_H)^\top \in \mathbb{R}^H$ is a linear filter.

For our constrained model, an unrestricted filter is insufficient: its action can violate (2) by exceeding αx_t or becoming negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction ensures that every fixed policy is feasible while preserving convex surrogate losses and tractable online updates.

B. SDAC Definition

Fix a memory horizon $H \in \mathbb{N}_{>0}$. Let

$$\mathcal{M}_H \triangleq \left\{\mathbf{m} \in \mathbb{R}^H_{\ge 0} : \|\mathbf{m}\|_1 \le 1\right\} \tag{9}$$

be the subprobability simplex. For any $\mathbf{m} \in \mathcal{M}_H$, define the policy $\pi_{\mathbf{m}}$ by

$$u_t(\mathbf{m}) \;\triangleq\; \sum_{i=1}^H m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle, \tag{10}$$

where

$$\boldsymbol{\phi}_t^u \triangleq \big(\alpha w_{t-1}, \alpha^2 w_{t-2}, \ldots, \alpha^H w_{t-H}\big)^\top \in \mathbb{R}^H, \tag{11}$$

The induced state trajectory satisfies

$$x_{t+1}(\mathbf{m}) = \alpha x_t(\mathbf{m}) - u_t(\mathbf{m}) + w_t, \qquad x_1(\mathbf{m}) = 0. \tag{12}$$

The SDAC policy class is

$$\Pi_H^{\mathrm{SDAC}} \triangleq \{\pi_{\mathbf{m}} : \mathbf{m} \in \mathcal{M}_H\}. \tag{13}$$

The weights α^i align the arrival filter with the retention coefficient α in (3).

We now establish the two structural properties that make SDAC useful: the induced control action always satisfies the feasibility constraint (2), and the induced cost is convex and Lipschitz in $\mathbf{m}$, so that regret minimization against the best fixed SDAC policy is an instance of online convex optimization over the simplex. Both properties follow from an explicit decomposition of the state trajectory that isolates its dependence on $\mathbf{m}$.

Lemma 1 (State decomposition under SDAC). *Fix a positive integer H and $\mathbf{m} \in \mathcal{M}_H$. For $i \in \{1,\ldots,H\}$, define*

$$r_t(i) \triangleq \sum_{j=1}^i \alpha^{j-1} w_{t-j},$$

where the sum is empty when $i=0$, and write $r_t \triangleq r_t(t-1)$. Then, for every $t \in [T]$,

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_t(i) + \big(1 - \sum_{i=1}^{H} m_i\big) r_t. \tag{14}$$

Define the quantity

$$\boldsymbol{\phi}_t^x \triangleq \big(r_t(1) - r_t, \ldots, r_t(H) - r_t\big)^\top.$$

Then

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle.$$

Moreover, for all $i \in \{1, \ldots, H\}$,

$$0 \le r_t(i) \le r_t \le \frac{1}{1-\alpha},$$

and $\mathbf{m} \mapsto x_t(\mathbf{m})$ is affine.

The proof is given in Appendix B.

The first consequence of this decomposition is that every fixed policy is feasible.

Lemma 2 (Feasibility of SDAC policies)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the trajectory induced by* (10) *and* (12) *satisfies the action constraint* (2)*; explicitly,*

$$0 \le u_t(\mathbf{m}) \le \alpha x_t(\mathbf{m}) \qquad \textit{for all } t \in [T]. \tag{15}$$

Proof. Because m_i, α, and w_{t-i} are nonnegative,

$$u_t(\mathbf{m}) = \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \ge 0.$$

For every $i \in \{1, \ldots, H\}$,

$$\alpha r_t(i) = \sum_{j=1}^{i} \alpha^j w_{t-j} \ge \alpha^i w_{t-i}. \tag{16}$$

Because $\mathbf{m} \in \mathcal{M}_H$ and $r_t \ge 0$ by Lemma 1,

$$\begin{aligned} \alpha x_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha r_t(i) + \alpha\big(1 - \sum_{i=1}^{H} m_i\big) r_t \\ &\ge \sum_{i=1}^{H} m_i \alpha r_t(i) \\ &\ge \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = u_t(\mathbf{m}), \end{aligned} \tag{17}$$

where the second inequality uses (16). □

For fixed arrivals, both coordinates of the SDAC state–action pair are affine in $\mathbf{m}$. The following lemma records the resulting convexity and the constants used by the online update.

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H,$$

and

$$G_\alpha \triangleq \alpha \left(\frac{L_x}{1-\alpha} + L_u \right). \tag{18}$$

Then ℓ_t is convex on $\mathcal{M}_H$, and for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \le G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.$$

Moreover, at every $\mathbf{m} \in \mathcal{M}_H$, one can select $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$ such that $\|\mathbf{g}_t\|_\infty \le G_\alpha$.

For the online update, let (ζ_t^x, ζ_t^u) be the bounded cost subgradient at the induced state–action pair and use the explicit choice

$$\mathbf{g}_t = \zeta_t^x \boldsymbol{\phi}_t^x + \zeta_t^u \boldsymbol{\phi}_t^u$$

This choice belongs to $\partial \ell_t(\mathbf{m})$ and satisfies $\|\mathbf{g}_t\|_\infty \le G_\alpha$, as stated in Lemma 3. The proof is given in Appendix B.

Regret minimization against the best fixed SDAC policy thus becomes online convex optimization over the simplex.

C. Expressiveness: Decreasing-Allocation Policies

Energy-harvesting policies for Bernoulli recharges can be described by the sequence of expenditures following each recharge [11], [12]. In the corresponding decreasing-allocation policy class, each unit arriving at time τ is assigned the same decreasing allocation sequence, beginning at time $\tau+1$ to respect the information pattern. The sequence may have an arbitrary decreasing shape; geometric sequences induce proportional state-feedback policies.

Define the set of feasible decreasing allocation sequences

$$\mathcal{Q}_\alpha^{\downarrow} \triangleq \left\{ \mathbf{q} = (q_i)_{i \ge 1} : q_1 \ge q_2 \ge \cdots \ge 0, \quad \sum_{i=1}^{\infty} \frac{q_i}{\alpha^i} \le 1 \right\}. \tag{19}$$

For $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$, define

$$\begin{aligned} u_t(\mathbf{q}) &\triangleq \sum_{i=1}^{t-1} q_i w_{t-i}, \\ x_{t+1}(\mathbf{q}) &= \alpha x_t(\mathbf{q}) - u_t(\mathbf{q}) + w_t, \qquad x_1(\mathbf{q}) = 0. \end{aligned} \tag{20}$$

Here $q_i w_s$ is the amount from arrival w_s allocated i periods later. The weighted budget in (19) accounts for leakage: reserving q_i units for expenditure i periods after an arrival requires q_i/α^i units at the arrival time. Let

$$\Pi_\alpha^{\mathrm{alloc}} \triangleq \{\pi_{\mathbf{q}} : \mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}\}.$$

Lemma 4 (Finite-memory approximation of decreasing-allocation policies)**.** *Every policy in $\Pi_\alpha^{\mathrm{alloc}}$ is feasible. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$ and $H \ge 1$, and define*

$$m_{H,i}(\mathbf{q}) \triangleq \frac{q_i}{\alpha^i}, \qquad i \in [H]. \tag{21}$$

Then $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. With

$$\mathrm{Tail}_H(\mathbf{q}) \triangleq \sum_{i=H+1}^{\infty} q_i,$$

the corresponding trajectories satisfy, for every arrival sequence in $[0,1]$,

$$0 \le u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) \le \mathrm{Tail}_H(\mathbf{q}), \tag{22}$$

$$0 \le x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}) \le \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha}. \tag{23}$$

If each c_t satisfies (5)*, then*

$$\left|\sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) - \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\right| \le \left(L_u + \frac{L_x}{1-\alpha}\right) \mathrm{Tail}_H(\mathbf{q}) T \le \left(L_u + \frac{L_x}{1-\alpha}\right) \alpha^{H+1} T. \tag{24}$$

Corollary 1 (Proportional state feedback as a geometric allocation)**.** *Fix $k \in [0, \alpha]$, set $\lambda_k \triangleq \alpha - k$, and define*

$$q_i(k) = k\lambda_k^{i-1}, \qquad i \ge 1, \tag{25}$$

where $\lambda_k^0 \triangleq 1$. Then $\mathbf{q}(k) \in \mathcal{Q}_\alpha^{\downarrow}$ and $\pi_{\mathbf{q}(k)}$ is exactly the feasible proportional state-feedback policy $u_t(k) = kx_t(k)$. Its SDAC truncation is

$$\mathbf{m}_H^{\mathrm{prop}}(k) \triangleq \mathbf{m}_H\big(\mathbf{q}(k)\big),$$

with coordinates

$$m_{H,i}\big(\mathbf{q}(k)\big) = \frac{k\lambda_k^{i-1}}{\alpha^i}, \qquad i \in [H],$$

and its tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = \frac{k\lambda_k^H}{1-\lambda_k}.$$

Therefore, (24) *specializes exactly to*

$$\left|\sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}_H^{\mathrm{prop}}(k)), u_t(\mathbf{m}_H^{\mathrm{prop}}(k))\big) - \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)\right| \le \left(L_u + \frac{L_x}{1-\alpha}\right) \frac{k\lambda_k^H}{1-\lambda_k} T. \tag{26}$$

The proofs of Lemma 4 and Corollary 1 are given in Appendix B.

Thus, $\Pi_\alpha^{\mathrm{alloc}}$ contains every feasible proportional state-feedback policy and also permits nongeometric allocation sequences. Together with Theorem 1, the tail bound in (24) yields no regret against $\Pi_\alpha^{\mathrm{alloc}}$ and, consequently, against feasible proportional state-feedback policies.

V. Online Algorithm

We now present our online algorithm, which updates SDAC parameters by OMD with unnormalized negentropy (equivalently, the exponentiated gradient method) and a fixed step size $\eta > 0$ over the subprobability simplex $\mathcal{M}_H \subset \mathbb{R}^H$ [26]. Let

$$\Psi(\mathbf{m}) \triangleq \sum_{i=1}^{H} \big(m_i \log m_i - m_i\big), \qquad \mathbf{m} \in \mathbb{R}_{\ge 0}^H, \tag{27}$$

denote the unnormalized negentropy (with the convention $0 \log 0 \triangleq 0$). Its Bregman divergence is the generalized Kullback–Leibler divergence

$$D_\Psi(\mathbf{m}\|\mathbf{v}) \triangleq \sum_{i=1}^{H} \Big(m_i \log \tfrac{m_i}{v_i} - m_i + v_i\Big), \tag{28}$$

defined for

$$\mathbf{m} \in \mathbb{R}_{\ge 0}^H, \qquad \mathbf{v} \in \mathbb{R}_{>0}^H.$$

For each time $t \in [T]$, let $\mathbf{m}_t \in \mathcal{M}_H$ be the online parameter. We initialize at

$$\mathbf{m}_1 \triangleq \frac{1}{H+1}\mathbf{1} \tag{29}$$

so that $\sup_{\mathbf{m} \in \mathcal{M}_H} D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \log(H+1)$ (see the proof of Theorem 1).

a) Control-action selection: The controller first forms the nominal SDAC action

$$\widehat{u}_t \triangleq \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t \rangle, \tag{30}$$

and clips it to the admissible interval in (2):

$$u_t \triangleq \min\{\widehat{u}_t, \alpha x_t\}. \tag{31}$$

The state then evolves according to (3). Because $\widehat{u}_t \ge 0$, the clipping rule and induction on (3), starting from (1) with $w_t \in [0,1]$, yield $x_t \ge 0$ for all $t \in [T+1]$ and show that the online trajectory satisfies (2).

The scalar correction in (31) also has an exact Bregman-projection interpretation. Lemma 6 shows that the negentropy projection of $\mathbf{m}_t$ onto the parameters compatible with the available state induces exactly the clipped action u_t.

This interpretation does not change the implementation. The full H-dimensional execution vector $\mathbf{m}_t^{\mathrm{exec}}$ is never calculated because only its scalar induced action is executed. In particular, the OMD update below uses the internal iterate $\mathbf{m}_t$, not $\mathbf{m}_t^{\mathrm{exec}}$. Therefore the internal iterates remain on the fixed domain $\mathcal{M}_H$, and the analysis does not acquire a time-varying comparator set. Forming the feature vectors and their inner products, followed by the OMD update, takes $O(H)$ time and $O(H)$ memory per round.

b) Surrogate cost and OMD update: After (w_t, c_t) is revealed, form the surrogate subgradient displayed after Lemma 3, $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$, which satisfies $\|\mathbf{g}_t\|_\infty \le G_\alpha$ with G_α defined in (18). This step uses the revealed cost function and the full-information subgradient oracle from Section III. We define the OMD update

$$\mathbf{m}_{t+1} \triangleq \operatorname*{arg\,min}_{\mathbf{m} \in \mathcal{M}_H} \Big\{\eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\Big\}. \tag{32}$$

c) Closed-form solution: Although (32) is a constrained optimization problem, it is easy to solve analytically because Ψ is separable and the objective is linear in $\mathbf{g}_t$. As shown in Lemma 8, the minimizer in (32) is exactly the result of two elementary steps. First a multiplicative step,

$$\widetilde{m}_{t+1,i} \triangleq m_{t,i} \exp\big(-\eta g_{t,i}\big), \qquad i = 1, \ldots, H, \tag{33}$$

where $g_{t,i}$ is the i-th coordinate of $\mathbf{g}_t$. This is followed by the Bregman projection onto $\mathcal{M}_H$,

$$\mathbf{m}_{t+1} \triangleq \mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}_{t+1}) = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}, \tag{34}$$

where the projection operator is defined and derived in Lemma 7 of Appendix C; the scaling is nontrivial only when $\|\widetilde{\mathbf{m}}_{t+1}\|_1 > 1$.

VI. Regret Analysis

The following theorem states that the online controller achieves sublinear regret against every fixed SDAC policy $\pi_{\mathbf{m}}$, where $\mathbf{m}$ is any parameter in $\mathcal{M}_H$. The proof combines a standard online convex optimization bound for the surrogate costs ℓ_t with the trajectory-stability result stated below. The complete proofs are deferred to Appendix D.

Theorem 1 (Regret bound)**.** *Suppose the cost functions satisfy the regularity and bounded-subgradient assumptions of Section III. Then the OMD controller with action clipping from Section V, initialized as in* (29) *and using step size $\eta > 0$, satisfies, for the regret defined in* (7)*,*

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha, \tag{35}$$

where

$$\Gamma_\alpha \triangleq \frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}, \tag{36}$$

and $G_\alpha = \alpha\big(\frac{L_x}{1-\alpha} + L_u\big)$ *as in* (18)*. In particular, the choice*

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}$$

gives

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T \log(H+1)}.$$

For fixed $L_x, L_u > 0$, the leading factor obeys

$$\sqrt{\Gamma_\alpha} = \Theta(\alpha) \quad \text{as } \alpha \downarrow 0,$$

and

$$\sqrt{\Gamma_\alpha} = \Theta\Big((1-\alpha)^{-3/2}\Big) \quad \text{as } \alpha \uparrow 1.$$

The nonstandard step is to control the difference between the realized online trajectory and the fixed-policy trajectory associated with the current internal iterate.

Lemma 5 (Trajectory stability)**.** *For the online trajectory and the counterfactual fixed-policy trajectory associated with the current internal iterate, one has, for every $t \in [T]$,*

$$|x_t - x_t(\mathbf{m}_t)| \le \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \tag{37}$$

$$0 \le u_t(\mathbf{m}_t) - u_t \le \frac{\alpha^2\eta G_\alpha}{(1-\alpha)^2}. \tag{38}$$

Here $x_t(\mathbf{m}_t)$ is generated by using the single fixed parameter $\mathbf{m}_t$ from time 1 *through time t. Consequently,*

$$\sum_{t=1}^{T}\Big(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\Big) \le \frac{\alpha\eta T G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}. \tag{39}$$

The proof of Lemma 5 is given in Appendix D. The proof of Theorem 1 then has three steps. First, the OMD bound for the surrogate costs gives

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}) \le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}.$$

Second, (39) controls the difference between the realized and surrogate costs. Third, adding the two bounds and using (36) gives (35); optimizing over η gives the stated choice η^*.

Theorem 1 and Lemma 4 give the following comparator guarantee.

Corollary 2 (No-regret against decreasing-allocation policies)**.** *Set*

$$H_T \triangleq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil\right\},$$

and run the OMD controller with action clipping, memory horizon $H = H_T$, and the step size η^* of Theorem 1. For $\mathbf{q} \in \mathcal{Q}_\alpha^{\downarrow}$, let

$$J_T(\mathbf{q}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)$$

and

$$\mathrm{Reg}_T^{\mathrm{alloc}} \triangleq J_T^{\mathrm{on}} - \inf_{\mathbf{q}\in\mathcal{Q}_\alpha^{\downarrow}} J_T(\mathbf{q}).$$

Then

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{alloc}} &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1} T.\end{aligned} \tag{40}$$

Here Γ_α is defined in (36)*. Since $\alpha^{H_T} \le 1/T$ and $\log(H_T + 1) = O(\log\log T)$ as $T \to \infty$, the approximation term is $O(1)$. Consequently,*

$$\mathrm{Reg}_T^{\mathrm{alloc}} = O\Big(\sqrt{T\log\log T}\Big) = o(T).$$

Because geometric allocation sequences induce proportional state feedback, the same controller also satisfies a gain-dependent bound for that policy family.

Corollary 3 (No-regret against proportional state-feedback policies)**.** *Run the controller of Corollary 2. For $k \in [0,\alpha]$, let $J_T(k) \triangleq \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)$, where $u_t(k) = kx_t(k)$, and define*

$$\mathrm{Reg}_T^{\mathrm{prop}} \triangleq J_T^{\mathrm{on}} - \min_{k\in[0,\alpha]} J_T(k).$$

For every $k \in [0,\alpha]$, with $\lambda_k = \alpha - k$,

$$\begin{aligned}J_T^{\mathrm{on}} - J_T(k) &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^{H_T}}{1-\lambda_k} T.\end{aligned} \tag{41}$$

Moreover,

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{prop}} &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1} T \\ &= O\Big(\sqrt{T\log\log T}\Big) = o(T).\end{aligned}$$

The proofs of Corollaries 2 and 3 are given in Appendix D.

VII. Minimax Lower Bound

The next result shows that the $\sqrt{T}$ dependence is unavoidable and quantifies the dependence on $1 - \alpha$ that any causal feasible controller must incur. Its proof is deferred to Appendix A.

Theorem 2 (Minimax SDAC regret lower bound)**.** *Fix $H \ge 1$ and $T \ge 2$. For every causal feasible online controller $\mathcal{A}$,* *possibly randomized, there exists a deterministic oblivious sequence of arrivals and costs satisfying the assumptions of Section III such that*

$$\mathbb{E}_{\mathcal{A}}\left[\mathrm{Reg}_T^{\mathrm{SDAC}}\right] \geq \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}, \tag{42}$$

where

$$\beta_t \triangleq \alpha\left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha}\right). \tag{43}$$

The expectation is only over the possible internal randomization of $\mathcal{A}$; the adversarial sequence whose existence is asserted is deterministic and fixed before the interaction.

Corollary 4 (Lower bound for sufficiently long horizons)**.** *Define*

$$t_{\mathrm{mix}}(\alpha) \triangleq \left\lceil \frac{\log 2}{\log(1/\alpha)} \right\rceil. \tag{44}$$

If $T \geq 2t_{\mathrm{mix}}(\alpha) + 2$, then every causal feasible controller $\mathcal{A}$ admits a deterministic oblivious sequence for which

$$\mathbb{E}_{\mathcal{A}}\left[\mathrm{Reg}_T^{\mathrm{SDAC}}\right] \geq \frac{G_\alpha}{8}\sqrt{T}, \tag{45}$$

where $G_\alpha = \alpha(L_x/(1-\alpha) + L_u)$ as in (18)*.*

Remark 1 (Large-α regime). As $\alpha \uparrow 1$, stored arrivals persist longer and $t_{\mathrm{mix}}(\alpha) = \Theta((1-\alpha)^{-1})$. Hence, Corollary 4 yields

$$\mathbb{E}_{\mathcal{A}}\left[\mathrm{Reg}_T^{\mathrm{SDAC}}\right] = \Omega\left(\frac{\sqrt{T}}{1-\alpha}\right) \tag{46}$$

whenever $T \geq 2t_{\mathrm{mix}}(\alpha) + 2$. The upper bound in Theorem 1 is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$, so a factor $(1-\alpha)^{-1/2}$ between the two bounds remains open.

The condition $T \geq 2t_{\mathrm{mix}}(\alpha)+2$ is essential for interpreting the large-α asymptotic. If T is fixed while $\alpha \uparrow 1$, then

$$\frac{1-\alpha^{t-2}}{1-\alpha} \longrightarrow t-2,$$

and the right-hand side of (42) converges to the finite quantity

$$\frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\big(L_u + L_x(t-2)\big)^2}.$$

Thus, for fixed T, the exact finite-horizon bound in (42) applies instead of the long-horizon scaling in (46).

VIII. Discussion and Extensions

We briefly discuss three extensions. In each case, the feasibility, convexity, and stability arguments used above continue to apply after the corresponding changes to the policy domain and the Lipschitz constants. We state these changes but omit the resulting regret bounds, whose form is unchanged up to the modified constants and, in the vector settings, the chosen product norm.

a) Retention $\alpha \geq 1$.: For $\alpha \geq 1$, fix k such that $\bar{\alpha} \triangleq \alpha - k \in (0,1)$ and write $u_t = kx_t + v_t$. Then

$$x_{t+1} = \bar{\alpha}x_t - v_t + w_t.$$

An SDAC filter for v_t uses the powers $\bar{\alpha}^i$. If $0 \leq v_t \leq \bar{\alpha}x_t$, then

$$kx_t \leq u_t = kx_t + v_t \leq (k+\bar{\alpha})x_t = \alpha x_t,$$

so the original feasibility constraint (2) is satisfied. Since

$$x_{t+1} \leq \bar{\alpha}x_t + 1, \qquad x_1 = 0,$$

induction gives $x_t \leq 1/(1-\bar{\alpha})$. The transformed cost $c_t(x, kx+v)$ remains convex in (x,v). Thus the analysis applies when the costs satisfy the corresponding Lipschitz and bounded-subgradient assumptions on the bounded domain

$$0 \leq x \leq \frac{1}{1-\bar{\alpha}}, \qquad kx \leq u \leq \alpha x.$$

A remark is that for a fixed k, the comparator class is correspondingly restricted to policies whose actions satisfy $u_t \geq kx_t$. Selecting k (fixed or even in an online fashion) remains an interesting direction for future work.

b) Vector action / multi-task allocation.: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\geq 0}$ that allocates resources across n tasks, subject to $\sum_{a=1}^n u_{t,a} \leq \alpha x_t$, so that the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \sum_{a=1}^n u_{t,a} + w_t$, and the cost $c_t(x_t, \mathbf{u}_t)$ is convex in $(x, \mathbf{u})$. Replace $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n\times H}_{\geq 0}$ satisfying

$$\sum_{a=1}^{n}\sum_{i=1}^{H} M_{a,i} \leq 1, \qquad u_{t,a}(\mathbf{M}) = \sum_{i=1}^{H} M_{a,i}\alpha^i w_{t-i}.$$

Set $m_i = \sum_{a=1}^n M_{a,i}$. Then

$$m_i \geq 0, \qquad \sum_{i=1}^{H} m_i = \sum_{a=1}^{n}\sum_{i=1}^{H} M_{a,i} \leq 1,$$

so $\mathbf{m} \in \mathcal{M}_H$. Moreover,

$$\sum_{a=1}^{n} u_{t,a}(\mathbf{M}) = \sum_{i=1}^{H} m_i\alpha^i w_{t-i} \leq \alpha x_t(\mathbf{M}),$$

where the inequality is the scalar SDAC feasibility bound. The state and vector action remain affine in $\mathbf{M}$, so $c_t(x_t(\mathbf{M}), \mathbf{u}_t(\mathbf{M}))$ is convex in $\mathbf{M}$. For the online controller, a nominal vector that exceeds the available total resource can be scaled proportionally so that its entries remain nonnegative and sum to αx_t. Under an ℓ_1 Lipschitz bound in the action, the size of this correction equals the excess total action, and the stability proof proceeds with the corresponding matrix norm. OMD then runs over the matrix subprobability simplex above.

c) Multiple storages with diagonal dynamics.: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\geq 0}$ and evolves as $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$ with $\mathbf{A} = \mathrm{diag}(\alpha_1, \ldots, \alpha_d)$, $\alpha_s \in (0,1)$ for $s \in [d]$, arrivals $\mathbf{w}_t \in [0,1]^d$, and per-storage constraints $0 \leq u_{t,s} \leq \alpha_s x_{t,s}$. Assign one SDAC parameter in $\mathcal{M}_H$ to each storage and use the powers α_s^i in coordinate s. The dynamics and feasibility constraints then decouple by storage, and the fixed-policy domain is a product of subprobability simplices, with

$x_{t,s} \le 1/(1-\alpha_s)$ for every $s \in [d]$. The online controller clips each nominal action to $\alpha_s x_{t,s}$. If the cost is jointly convex and satisfies compatible Lipschitz and bounded-subgradient assumptions on the resulting bounded product domain, the surrogate cost is jointly convex in all SDAC parameters and OMD applies on the product set. The update separates by storage only when the cost is coordinate-separable; otherwise it is a joint update. Cross-coupled storage systems (non-diagonal $\mathbf{A}$) require additional analysis and remain an interesting direction for future work.

IX. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. We constructed the SDAC so that every policy trajectory is feasible, and an online algorithm with $O\left(\sqrt{T\log(H+1)}\right)$ regret against the best fixed SDAC policy. Once T is at least on the order of $(1-\alpha)^{-1}$, the minimax lower bound gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ regret as $\alpha \uparrow 1$; hence a factor $(1-\alpha)^{-1/2}$ between the upper and lower bounds remains open. The SDAC approximation results further give $O\left(\sqrt{T\log\log T}\right)$ regret against the best decreasing-allocation policy when $H = \Theta(\log T)$. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time.

Appendix A
Proofs for the Minimax Lower Bound

Proof of Theorem 2. Fix a causal feasible controller $\mathcal{A}$. Let $\xi_1, \ldots, \xi_T$ be independent Rademacher random variables, also independent of the internal randomization of $\mathcal{A}$, with

$$\mathbb{P}(\xi_t = 1) = \mathbb{P}(\xi_t = -1) = \frac{1}{2}.$$

Set

$$w_t = 1 \qquad \text{for all } t \in [T], \tag{47}$$

and define the costs by

$$c_t(x,u) = \xi_t(L_x x - L_u u) \qquad \text{for all } t \in [T]. \tag{48}$$

Each c_t is affine and hence convex, and, for any two points in $\mathcal{D}_\alpha$,

$$\begin{aligned} |c_t(x,u) - c_t(x',u')| &= |L_x(x-x') - L_u(u-u')| \\ &\le L_x|x-x'| + L_u|u-u'|. \end{aligned} \tag{49}$$

Moreover,

$$\nabla c_t = \xi_t(L_x, -L_u), \qquad |\partial_x c_t| = L_x, \qquad |\partial_u c_t| = L_u,$$

so the required subgradient bounds hold.

We now select two benchmark policies. The parameter $\mathbf{m}^{(0)}$ leaves all mass in the subprobability simplex unused, so its SDAC action is always zero. The parameter $\mathbf{m}^{(1)}$ places all mass on the first lag, so its action is αw_{t-1}. Let

$$\mathbf{m}^{(0)} \triangleq \mathbf{0}, \qquad \mathbf{m}^{(1)} \triangleq \mathbf{e}_1 \tag{50}$$

belong to $\mathcal{M}_H$, where $\mathbf{e}_1 = (1,0,\ldots,0) \in \mathbb{R}^H$. Thus, under (47), $\mathbf{m}^{(0)}$ is the no-release policy, whereas $\mathbf{m}^{(1)}$ releases the retained portion of the preceding arrival. For the no-release policy,

$$u_t(\mathbf{m}^{(0)}) = 0, \qquad x_t(\mathbf{m}^{(0)}) = \rho_t \triangleq \frac{1-\alpha^{t-1}}{1-\alpha}. \tag{51}$$

This follows by unrolling $x_{t+1} = \alpha x_t + 1$:

$$x_t(\mathbf{m}^{(0)}) = \sum_{s=0}^{t-2} \alpha^s = \frac{1-\alpha^{t-1}}{1-\alpha}, \qquad t \ge 2,$$

and the closed form also gives $x_1(\mathbf{m}^{(0)}) = 0$. For the one-lag policy, (10) gives

$$u_1(\mathbf{m}^{(1)}) = 0, \qquad u_t(\mathbf{m}^{(1)}) = \alpha, \quad t \ge 2. \tag{52}$$

Its state at $t = 2$ is

$$x_2(\mathbf{m}^{(1)}) = \alpha x_1(\mathbf{m}^{(1)}) - u_1(\mathbf{m}^{(1)}) + w_1 = 1.$$

If $x_t(\mathbf{m}^{(1)}) = 1$ for some $t \ge 2$, then

$$\begin{aligned} x_{t+1}(\mathbf{m}^{(1)}) &= \alpha x_t(\mathbf{m}^{(1)}) - u_t(\mathbf{m}^{(1)}) + w_t \\ &= \alpha - \alpha + 1 = 1. \end{aligned}$$

Induction therefore gives

$$x_t(\mathbf{m}^{(1)}) = 1, \qquad t \ge 2. \tag{53}$$

For $t \ge 2$, the difference between the two states is

$$\begin{aligned} \Delta_t &\triangleq x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)}) \\ &= \rho_t - 1 \\ &= \frac{\alpha(1-\alpha^{t-2})}{1-\alpha}. \end{aligned} \tag{54}$$

The two policies have the same state and action at $t = 1$. For $t \ge 2$, (52) and (54) give

$$J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)}) = \sum_{t=2}^{T} \xi_t(L_x\Delta_t + \alpha L_u) = \sum_{t=2}^{T} \xi_t\beta_t, \tag{55}$$

with β_t as in (43).

Let $\mathcal{F}_t^-$ be the information available immediately before c_t is revealed, including the randomization used to select u_t. By causality, (x_t, u_t) is $\mathcal{F}_t^-$-measurable, whereas ξ_t is independent of $\mathcal{F}_t^-$ and has mean zero. Hence, by the tower property,

$$\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}}] = \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\left[(L_x x_t - L_u u_t)\mathbb{E}[\xi_t \mid \mathcal{F}_t^-]\right] = 0. \tag{56}$$

The fixed-policy trajectories depend only on the arrival sequence and are therefore independent of the Rademacher variables. Thus, for $j \in \{0,1\}$,

$$\mathbb{E}_{\boldsymbol{\xi}}\left[J_T(\mathbf{m}^{(j)})\right] = \sum_{t=1}^{T} \mathbb{E}[\xi_t]\left(L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)})\right) = 0. \tag{57}$$

Because $\mathbf{m}^{(0)}, \mathbf{m}^{(1)} \in \mathcal{M}_H$, for every realization of the Rademacher variables and every choice made by the controller,

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{SDAC}} &= J_T^{\mathrm{on}} - \min_{\mathbf{m}\in\mathcal{M}_H} J_T(\mathbf{m}) \\ &\ge J_T^{\mathrm{on}} - \min\left\{J_T(\mathbf{m}^{(0)}), J_T(\mathbf{m}^{(1)})\right\}. \end{aligned} \tag{58}$$

Using $\min\{a,b\} = (a+b-|a-b|)/2$ together with (56) and (57), and then taking expectations in (58), gives

$$\begin{aligned}\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] &\geq \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left[\left|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\right|\right] \\ &= \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left[\left|\sum_{t=2}^{T}\xi_t\beta_t\right|\right] \\ &\geq \frac{1}{2\sqrt{2}}\left(\sum_{t=2}^{T}\beta_t^2\right)^{1/2}.\end{aligned} \tag{59}$$

The last step follows from Khintchine's inequality for Rademacher sums.

It remains to remove the randomization from the adversarial construction. The left-hand side of (59) is the average over all 2^T realizations of the Rademacher vector $\boldsymbol{\xi}$:

$$\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] = \frac{1}{2^T}\sum_{\boldsymbol{\xi}\in\{-1,1\}^T}\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\,\big|\,\boldsymbol{\xi}\big].$$

Hence at least one vector $\boldsymbol{\xi}^\star$ makes the conditional expectation at least the right-hand side of (59). Fixing $\boldsymbol{\xi}^\star$ in (48), together with the arrivals in (47), produces a deterministic admissible sequence fixed before the interaction. The remaining expectation is only over the internal randomization of $\mathcal{A}$, and (42) follows. □

Proof of Corollary 4. The definition of $t_{\mathrm{mix}}(\alpha)$ gives

$$\alpha^{t_{\mathrm{mix}}(\alpha)} \leq \alpha^{\log 2/\log(1/\alpha)} = \frac{1}{2}. \tag{60}$$

Therefore, for every $t \geq t_{\mathrm{mix}}(\alpha)+2$,

$$\begin{aligned}\beta_t &\geq \alpha\left(L_u + \frac{L_x}{2(1-\alpha)}\right) \\ &\geq \frac{\alpha}{2}\left(L_u + \frac{L_x}{1-\alpha}\right) = \frac{G_\alpha}{2}.\end{aligned} \tag{61}$$

There are $T - t_{\mathrm{mix}}(\alpha) - 1$ indices in $\{t_{\mathrm{mix}}(\alpha)+2,\dots,T\}$. If $T \geq 2t_{\mathrm{mix}}(\alpha)+2$, then

$$T - t_{\mathrm{mix}}(\alpha) - 1 \geq \frac{T}{2}. \tag{62}$$

Applying Theorem 2 and retaining only these indices,

$$\begin{aligned}\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] &\geq \frac{1}{2\sqrt{2}}\left[\frac{T}{2}\left(\frac{G_\alpha}{2}\right)^2\right]^{1/2} \\ &= \frac{G_\alpha}{8}\sqrt{T}.\end{aligned} \tag{63}$$

This is (45). □

Appendix B
Proofs for Simplex Disturbance-Action Control

We collect the deferred proofs of the structural and expressiveness results from Section IV.

Proof of Lemma 1. The auxiliary quantities satisfy

$$r_{t+1} = \alpha r_t + w_t, \qquad r_1 = 0, \tag{64}$$

and, for every $i \in \{1,\dots,H\}$,

$$\begin{aligned}\alpha r_t(i) - \alpha^i w_{t-i} &= \sum_{j=1}^{i-1}\alpha^j w_{t-j} \\ &= r_{t+1}(i) - w_t.\end{aligned} \tag{65}$$

For $i=1$, the sum in the first line is empty.

At $t=1$, since $w_\tau \equiv 0$ for $\tau \leq 0$,

$$\begin{aligned}r_1(i) &= \sum_{j=1}^{i}\alpha^{j-1}w_{1-j} = 0, \\ r_1 &= 0, \\ x_1(\mathbf{m}) &= 0,\end{aligned}$$

so (14) holds. Suppose it holds at time t. Using (12) and (10),

$$\begin{aligned}x_{t+1}(\mathbf{m}) &= \alpha\left(\sum_{i=1}^{H}m_i r_t(i) + \big(1-\sum_{i=1}^{H}m_i\big)r_t\right) \\ &\quad - \sum_{i=1}^{H}m_i\alpha^i w_{t-i} + w_t \\ &= \sum_{i=1}^{H}m_i\big(\alpha r_t(i) - \alpha^i w_{t-i}\big) \\ &\quad + \alpha\big(1-\sum_{i=1}^{H}m_i\big)r_t + w_t.\end{aligned} \tag{66}$$

Substituting (64) and (65) into (66),

$$\begin{aligned}x_{t+1}(\mathbf{m}) &= \sum_{i=1}^{H}m_i\big(r_{t+1}(i) - w_t\big) \\ &\quad + \big(1-\sum_{i=1}^{H}m_i\big)(r_{t+1}-w_t) + w_t \\ &= \sum_{i=1}^{H}m_i r_{t+1}(i) + \big(1-\sum_{i=1}^{H}m_i\big)r_{t+1},\end{aligned} \tag{67}$$

because the total coefficient of w_t in the first line is $-\sum_{i=1}^{H}m_i - \big(1-\sum_{i=1}^{H}m_i\big) + 1 = 0$. This proves (14) by induction.

All summands defining $r_t(i)$ and r_t are nonnegative. Since $w_\tau = 0$ for $\tau \leq 0$, $r_t(i)$ is a truncation of r_t; hence

$$0 \leq r_t(i) \leq r_t \leq \sum_{j=1}^{\infty}\alpha^{j-1} = \frac{1}{1-\alpha}.$$

Finally,

$$x_t(\mathbf{m}) = r_t + \sum_{i=1}^{H}m_i\big(r_t(i) - r_t\big) = r_t + \langle\boldsymbol{\phi}_t^x, \mathbf{m}\rangle,$$

which is affine in $\mathbf{m}$. □

Proof of Lemma 3. Lemma 1 and (10) give

$$x_t(\mathbf{m}) = r_t + \langle\boldsymbol{\phi}_t^x, \mathbf{m}\rangle, \qquad u_t(\mathbf{m}) = \langle\boldsymbol{\phi}_t^u, \mathbf{m}\rangle. \tag{68}$$

For every $i \in \{1,\dots,H\}$, the definitions of $r_t(i)$ and r_t give

$$\begin{aligned}\phi_{t,i}^x &= r_t(i) - r_t \\ &= -\sum_{j=i+1}^{t-1}\alpha^{j-1}w_{t-j},\end{aligned} \tag{69}$$

where the sum is empty when $i \geq t-1$. Hence

$$\begin{aligned} \left|\phi^x_{t,i}\right| &\leq \sum_{j=i+1}^{\infty} \alpha^{j-1} = \frac{\alpha^i}{1-\alpha}, \\ 0 \leq \phi^u_{t,i} &= \alpha^i w_{t-i} \leq \alpha^i. \end{aligned} \tag{70}$$

Taking the largest coordinate, attained in these upper bounds at $i=1$, yields

$$\|\boldsymbol{\phi}^x_t\|_\infty \leq \frac{\alpha}{1-\alpha}, \qquad \|\boldsymbol{\phi}^u_t\|_\infty \leq \alpha. \tag{71}$$

Since (68) is affine in $\mathbf{m}$ and c_t is convex, ℓ_t is convex.

By (5) and (68),

$$\begin{aligned} |\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| &\leq L_x\left|\langle \boldsymbol{\phi}^x_t, \mathbf{m}-\mathbf{m}'\rangle\right| + L_u\left|\langle \boldsymbol{\phi}^u_t, \mathbf{m}-\mathbf{m}'\rangle\right| \\ &\leq \big(L_x\|\boldsymbol{\phi}^x_t\|_\infty + L_u\|\boldsymbol{\phi}^u_t\|_\infty\big)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &\leq \alpha\left(\frac{L_x}{1-\alpha} + L_u\right)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &= G_\alpha\|\mathbf{m}-\mathbf{m}'\|_1. \end{aligned} \tag{72}$$

Let (ζ^x_t, ζ^u_t) denote the components of the bounded subgradient $\boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$ from Section III, and set

$$\mathbf{g}_t \triangleq \zeta^x_t\, \boldsymbol{\phi}^x_t + \zeta^u_t\, \boldsymbol{\phi}^u_t. \tag{73}$$

For every $\mathbf{m}' \in \mathcal{M}_H$, the subgradient inequality and (68) yield

$$\begin{aligned} \ell_t(\mathbf{m}') &\geq \ell_t(\mathbf{m}) + \zeta^x_t\langle \boldsymbol{\phi}^x_t, \mathbf{m}'-\mathbf{m}\rangle + \zeta^u_t\langle \boldsymbol{\phi}^u_t, \mathbf{m}'-\mathbf{m}\rangle \\ &= \ell_t(\mathbf{m}) + \langle \mathbf{g}_t, \mathbf{m}'-\mathbf{m}\rangle. \end{aligned}$$

Thus $\mathbf{g}_t \in \partial\ell_t(\mathbf{m})$. For each $i \in \{1,\ldots,H\}$, the exact coordinate decay in (70) gives

$$\begin{aligned} |g_{t,i}| &\leq |\zeta^x_t|\left|\phi^x_{t,i}\right| + |\zeta^u_t|\phi^u_{t,i} \\ &\leq \alpha^i\left(\frac{L_x}{1-\alpha} + L_u\right) \\ &\leq \alpha\left(\frac{L_x}{1-\alpha} + L_u\right) = G_\alpha. \end{aligned} \tag{74}$$

Taking the maximum over i proves

$$\|\mathbf{g}_t\|_\infty \leq G_\alpha. \tag{75}$$

□

Proof of Lemma 4. Fix $\mathbf{q} \in \mathcal{Q}^\downarrow_\alpha$ and set

$$a_i \triangleq \frac{q_i}{\alpha^i}, \qquad b_j \triangleq 1 - \sum_{i=1}^{j-1} a_i, \qquad j \geq 1.$$

The weighted-budget constraint gives $a_i \geq 0$ and $b_j \geq 0$. Moreover,

$$x_t(\mathbf{q}) = \sum_{j=1}^{t-1} \alpha^{j-1} b_j w_{t-j}. \tag{76}$$

Indeed, the identity is immediate at $t=1$. If it holds at time t, then

$$\begin{aligned} x_{t+1}(\mathbf{q}) &= w_t + \sum_{j=1}^{t-1}\big(\alpha^j b_j - q_j\big)w_{t-j} \\ &= w_t + \sum_{j=1}^{t-1} \alpha^j b_{j+1} w_{t-j}, \end{aligned}$$

because $q_j = \alpha^j a_j$ and $b_{j+1} = b_j - a_j$. Reindexing gives (76) at time $t+1$.

For each $j \geq 1$,

$$a_j \leq 1 - \sum_{i=1}^{j-1} a_i = b_j.$$

Consequently,

$$\begin{aligned} 0 \leq u_t(\mathbf{q}) &= \sum_{j=1}^{t-1} \alpha^j a_j w_{t-j} \\ &\leq \sum_{j=1}^{t-1} \alpha^j b_j w_{t-j} = \alpha x_t(\mathbf{q}), \end{aligned}$$

which proves (2) for the decreasing-allocation policy.

For the truncated vector in (21),

$$m_{H,i}(\mathbf{q}) = a_i \geq 0, \qquad \sum_{i=1}^{H} m_{H,i}(\mathbf{q}) \leq \sum_{i=1}^{\infty} a_i \leq 1.$$

Thus $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. Its SDAC action is

$$u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=1}^{\min\{H,t-1\}} q_i w_{t-i}.$$

It follows that

$$0 \leq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=H+1}^{t-1} q_i w_{t-i} \leq \mathrm{Tail}_H(\mathbf{q}),$$

where the middle sum is empty when $t \leq H+1$. This proves (22).

Define

$$e^x_t \triangleq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}), \qquad e^u_t \triangleq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big).$$

The two state recursions give

$$e^x_{t+1} = \alpha e^x_t + e^u_t, \qquad e^x_1 = 0.$$

Since $0 \leq e^u_t \leq \mathrm{Tail}_H(\mathbf{q})$,

$$0 \leq e^x_t = \sum_{\tau=1}^{t-1} \alpha^{t-1-\tau} e^u_\tau \leq \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha},$$

which proves (23). The Lipschitz property now yields

$$\begin{aligned} &\big|c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) \\ &\quad - c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\big| \\ &\quad \leq \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q}). \end{aligned}$$

Finally,

$$\begin{aligned} \mathrm{Tail}_H(\mathbf{q}) &= \sum_{i=H+1}^{\infty} \alpha^i a_i \\ &\leq \alpha^{H+1} \sum_{i=H+1}^{\infty} a_i \leq \alpha^{H+1}. \end{aligned}$$

Summing the per-period cost bound proves (24). □

Proof of Corollary 1. Fix $k \in [0, \alpha]$, let $\lambda_k = \alpha - k$, and let $\mathbf{q}(k)$ be defined by (25). The sequence is nonnegative and decreasing. If $k = 0$, then $\mathbf{q}(k) = \mathbf{0} \in \mathcal{Q}_\alpha^\downarrow$. If $k > 0$, then $\lambda_k/\alpha < 1$ and

$$\sum_{i=1}^{\infty} \frac{q_i(k)}{\alpha^i} = \frac{k}{\alpha} \sum_{j=0}^{\infty} \left(\frac{\lambda_k}{\alpha}\right)^j = \frac{k}{\alpha - \lambda_k} = 1,$$

so again $\mathbf{q}(k) \in \mathcal{Q}_\alpha^\downarrow$. Substituting $u_t(k) = kx_t(k)$ into the dynamics and unrolling gives

$$u_t(k) = k \sum_{i=1}^{t-1} \lambda_k^{i-1} w_{t-i} = \sum_{i=1}^{t-1} q_i(k) w_{t-i} = u_t\big(\mathbf{q}(k)\big).$$

The two policies therefore have identical action and state trajectories. Their SDAC truncations also coincide coordinatewise, and the omitted geometric tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = k \sum_{i=H+1}^{\infty} \lambda_k^{i-1} = \frac{k\lambda_k^H}{1-\lambda_k}.$$

Substitution in (24) proves (26). □

Appendix C
Proofs for the Online Algorithm

The following lemmas establish the Bregman-projection interpretation of scalar action clipping, the Bregman-projection formula, and the equivalent closed-form OMD update used in Section V.

Lemma 6 (Action clipping as a Bregman projection)**.** *At time t, define the parameter set compatible with the available state by*

$$\mathcal{F}_t(x_t) \triangleq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \le \alpha x_t\}. \tag{77}$$

For the negentropy Ψ *in* (27)*, let*

$$\mathbf{m}_t^{\mathrm{exec}} \in \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{F}_t(x_t)} D_\Psi(\mathbf{m}\|\mathbf{m}_t). \tag{78}$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} = u_t. \tag{79}$$

Thus the projection formulation and the scalar clipping rule induce exactly the same executed action.

Proof. The set $\mathcal{F}_t(x_t)$ is nonempty because it contains $\mathbf{0}$, and it is compact and convex. By Lemma 8, $\mathbf{m}_t$ is strictly positive coordinatewise. Hence $D_\Psi(\cdot\|\mathbf{m}_t)$ is strictly convex on $\mathcal{M}_H$ and is minimized uniquely at $\mathbf{m}_t$ over any feasible set containing $\mathbf{m}_t$. Consequently, if

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle \le \alpha x_t,$$

then $\mathbf{m}_t^{\mathrm{exec}} = \mathbf{m}_t$, which proves (79) in the feasible case.

Suppose instead that $\mathbf{m}_t$ is infeasible. We claim that the action constraint is active at $\mathbf{m}_t^{\mathrm{exec}}$. If it were slack, then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle < \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle. \tag{80}$$

For sufficiently small $\lambda \in (0,1)$, the point

$$\mathbf{m}_\lambda \triangleq (1-\lambda)\mathbf{m}_t^{\mathrm{exec}} + \lambda \mathbf{m}_t$$

would still belong to $\mathcal{F}_t(x_t)$. Strict convexity and $D_\Psi(\mathbf{m}_t\|\mathbf{m}_t) = 0$ would then give

$$\begin{aligned} D_\Psi(\mathbf{m}_\lambda\|\mathbf{m}_t) &< (1-\lambda) D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) \\ &< D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t), \end{aligned}$$

contradicting (78). Hence

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \alpha x_t$$

in the infeasible case, which completes the proof. □

Lemma 7 (Bregman projection onto $\mathcal{M}_H$)**.** *Let* D_Ψ *be the Bregman divergence* (28) *of the unnormalized negentropy* (27)*. Fix* $\widetilde{\mathbf{m}} \in \mathbb{R}_{>0}^H$*. Then the Bregman projection of* $\widetilde{\mathbf{m}}$ *onto* $\mathcal{M}_H$*,*

$$\mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}) \triangleq \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}),$$

is given by

$$\mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\max\{1, \|\widetilde{\mathbf{m}}\|_1\}}.$$

Proof. The projection objective $\mathbf{m} \mapsto D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}})$ is strictly convex, and, for $m_i > 0$,

$$\frac{\partial}{\partial m_i} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}) = \log \frac{m_i}{\widetilde{m}_i}. \tag{81}$$

Its unconstrained minimizer is therefore $\mathbf{m} = \widetilde{\mathbf{m}}$.

If $\|\widetilde{\mathbf{m}}\|_1 \le 1$, this minimizer belongs to $\mathcal{M}_H$, and hence

$$\mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}) = \widetilde{\mathbf{m}}.$$

Suppose $\|\widetilde{\mathbf{m}}\|_1 > 1$. The constraint $\|\mathbf{m}\|_1 \le 1$ is then active. Moreover, the minimizer has no zero coordinate: if $m_i = 0$, transferring an arbitrarily small amount of mass from a positive coordinate to coordinate i has directional derivative $-\infty$, because

$$\lim_{z\downarrow 0} \log \frac{z}{\widetilde{m}_i} = -\infty.$$

Thus the nonnegativity constraints are inactive. With $\nu \ge 0$ denoting the multiplier of $\|\mathbf{m}\|_1 \le 1$, stationarity and complementary slackness give

$$\log \frac{m_i}{\widetilde{m}_i} + \nu = 0, \qquad \|\mathbf{m}\|_1 = 1. \tag{82}$$

The first relation in (82) implies

$$m_i = \widetilde{m}_i \exp(-\nu).$$

Summing over i and using the second relation in (82),

$$1 = \|\widetilde{\mathbf{m}}\|_1 \exp(-\nu), \qquad \exp(-\nu) = \frac{1}{\|\widetilde{\mathbf{m}}\|_1}.$$

Hence $\nu = \log \|\widetilde{\mathbf{m}}\|_1 > 0$ and

$$\mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\|\widetilde{\mathbf{m}}\|_1}.$$

Combining the cases $\|\widetilde{\mathbf{m}}\|_1 \le 1$ and $\|\widetilde{\mathbf{m}}\|_1 > 1$ proves the stated formula. □

Lemma 8 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$, and let $\widetilde{\mathbf{m}}_{t+1}$ be defined coordinatewise by* (33)*, i.e. $\widetilde{m}_{t+1,i} = m_{t,i}\exp(-\eta g_{t,i})$. Then*

$$\begin{aligned}&\operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H}\Big\{\eta\langle\mathbf{g}_t,\mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\Big\}\\ &\quad= \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) = \mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}),\end{aligned}$$

where $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}$ is the Bregman projection of Lemma 7. Consequently, the OMD update (32) *coincides exactly with the closed-form OMD update* (33)–(34)*, and in particular $\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$. Thus the positive initialization* (29) *keeps every algorithm iterate strictly positive.*

Proof. Fix $\mathbf{m} \in \mathbb{R}^H_{\geq 0}$ and a coordinate i. Then

$$\begin{aligned}\eta g_{t,i}\, m_i + m_i \log\frac{m_i}{m_{t,i}} &= m_i\left(\eta g_{t,i} + \log\frac{m_i}{m_{t,i}}\right)\\ &= m_i \log\frac{m_i}{\widetilde{m}_{t+1,i}}.\end{aligned} \tag{83}$$

Substitution into the OMD objective gives

$$\begin{aligned}&\eta\langle\mathbf{g}_t,\mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\\ &\quad= D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^{H}\left(m_{t,i} - \widetilde{m}_{t+1,i}\right).\end{aligned} \tag{84}$$

The final sum is independent of $\mathbf{m}$. Hence the two optimization problems have the same minimizer, and Lemma 7 gives (34). Since $\mathbf{m}_t > 0$ coordinatewise, both the multiplicative step and the subsequent positive rescaling preserve strict positivity. □

Appendix D
Proofs for the Regret Analysis

The following lemmas establish the OMD and stability estimates used in the proof of Theorem 1. They rely on the state-decomposition, feasibility, and surrogate-cost lemmas established in Section IV (Lemmas 1, 2, and 3).

Lemma 9 (One-step movement of OMD)**.** *For the OMD update with step size $\eta > 0$ and subgradient $\mathbf{g}_t \in \partial\ell_t(\mathbf{m}_t)$,*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta\,\|\mathbf{g}_t\|_\infty.$$

In particular, since $\|\mathbf{g}_t\|_\infty \leq G_\alpha$, $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta G_\alpha$.

Proof. First-order optimality of (32), evaluated at the feasible point $\mathbf{m}_t$, gives

$$\begin{aligned}&\langle\eta\mathbf{g}_t + \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m}_t - \mathbf{m}_{t+1}\rangle \geq 0,\\ &D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) + D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1})\\ &\qquad\leq \eta\langle\mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &\qquad\leq \eta\|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.\end{aligned} \tag{85}$$

The second line uses the symmetric Bregman identity.

For every $\mathbf{z} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$ and $\mathbf{v} \in \mathbb{R}^H$,

$$\begin{aligned}\mathbf{v}^\top\nabla^2\Psi(\mathbf{z})\mathbf{v} &= \sum_{i=1}^{H}\frac{v_i^2}{z_i}\\ &\geq \frac{\|\mathbf{v}\|_1^2}{\sum_{i=1}^{H} z_i}\\ &\geq \|\mathbf{v}\|_1^2.\end{aligned} \tag{86}$$

The first inequality is Cauchy–Schwarz. The segment joining $\mathbf{m}_t$ and $\mathbf{m}_{t+1}$ lies in $\mathcal{M}_H \cap \mathbb{R}^H_{>0}$ by Lemma 8; integrating (86) along this segment yields

$$\begin{aligned}D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t) &\geq \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2,\\ D_\Psi(\mathbf{m}_t\|\mathbf{m}_{t+1}) &\geq \frac{1}{2}\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2.\end{aligned}$$

Combining these inequalities with (85),

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2 \leq \eta\|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.$$

The claim is immediate if $\mathbf{m}_{t+1} = \mathbf{m}_t$; otherwise divide by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$. □

Lemma 10 (OMD regret for the surrogate losses)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (32) *satisfies*

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \sum_{t=1}^{T}\ell_t(\mathbf{m}) \leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2. \tag{87}$$

Proof. Convexity of ℓ_t gives

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \langle\mathbf{g}_t, \mathbf{m}_t - \mathbf{m}\rangle\\ &= \langle\mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &\quad+ \langle\mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle.\end{aligned} \tag{88}$$

First-order optimality of (32), evaluated at $\mathbf{m} \in \mathcal{M}_H$, yields

$$\begin{aligned}\eta\langle\mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m}\rangle &\leq \langle\nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m} - \mathbf{m}_{t+1}\rangle\\ &= D_\Psi(\mathbf{m}\|\mathbf{m}_t) - D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})\\ &\quad- D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t).\end{aligned} \tag{89}$$

The equality is the three-point Bregman identity. Substituting (89) into (88),

$$\begin{aligned}\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{t+1})}{\eta}\\ &\quad+ \langle\mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle\\ &\quad- \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}.\end{aligned} \tag{90}$$

The strong-convexity calculation (86) and Hölder's inequality give

$$\begin{aligned}&\langle\mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle - \frac{D_\Psi(\mathbf{m}_{t+1}\|\mathbf{m}_t)}{\eta}\\ &\quad\leq \|\mathbf{g}_t\|_\infty\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta}\\ &\quad\leq \frac{\eta}{2}\|\mathbf{g}_t\|_\infty^2.\end{aligned} \tag{91}$$

The last inequality is Young's inequality. Combining (90) and (91), and then summing over t, gives

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t)-\sum_{t=1}^{T}\ell_t(\mathbf{m})\le\frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta}-\frac{D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})}{\eta}+\frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2.$$

Since $D_\Psi(\mathbf{m}\|\mathbf{m}_{T+1})\ge 0$, this proves (87). $\square$

Lemma 11 (Positive-part representation of the dynamics)**.** *For $\mathbf{m}\in\mathcal{M}_H$, define*

$$F_t(x;\mathbf{m})\triangleq[\alpha x-\langle\boldsymbol{\phi}_t^u,\mathbf{m}\rangle]_+ + w_t. \tag{92}$$

Then the online state and every fixed-policy state satisfy

$$x_{t+1}=F_t(x_t;\mathbf{m}_t),\qquad x_{t+1}(\mathbf{m})=F_t(x_t(\mathbf{m});\mathbf{m}). \tag{93}$$

Moreover, for all $x,x'\ge 0$ and $\mathbf{m},\mathbf{m}'\in\mathcal{M}_H$,

$$\begin{aligned}&|F_t(x;\mathbf{m})-F_t(x';\mathbf{m}')|\\&\quad\le|\alpha(x-x')-\langle\boldsymbol{\phi}_t^u,\mathbf{m}-\mathbf{m}'\rangle|\\&\quad\le\alpha|x-x'|+\|\boldsymbol{\phi}_t^u\|_\infty\|\mathbf{m}-\mathbf{m}'\|_1.\end{aligned} \tag{94}$$

Proof. For the online trajectory, (31) implies

$$\begin{aligned}\alpha x_t-u_t&=\alpha x_t-\min\{\langle\boldsymbol{\phi}_t^u,\mathbf{m}_t\rangle,\alpha x_t\}\\&=[\alpha x_t-\langle\boldsymbol{\phi}_t^u,\mathbf{m}_t\rangle]_+ .\end{aligned} \tag{95}$$

For a fixed $\mathbf{m}$, Lemma 2 gives

$$\alpha x_t(\mathbf{m})-\langle\boldsymbol{\phi}_t^u,\mathbf{m}\rangle\ge 0,$$

so inserting a positive part does not change its state update. This proves (93). Finally, the map $z\mapsto[z]_+$ is nonexpansive:

$$|[z]_+-[z']_+|\le|z-z'|.$$

Apply this inequality to the two arguments in (92), and then apply Hölder's inequality, to obtain (94). $\square$

Proof of Lemma 5. Define

$$d_t\triangleq|x_t-x_t(\mathbf{m}_t)|.$$

Since $x_1=x_1(\mathbf{m}_1)=0$, one has $d_1=0$. For $t\in\{1,\dots,T-1\}$, Lemma 11 gives

$$\begin{aligned}d_{t+1}&=|F_t(x_t;\mathbf{m}_t)-F_t(x_t(\mathbf{m}_{t+1});\mathbf{m}_{t+1})|\\&\le\alpha\,|x_t-x_t(\mathbf{m}_{t+1})|+\|\boldsymbol{\phi}_t^u\|_\infty\|\mathbf{m}_{t+1}-\mathbf{m}_t\|_1.\end{aligned} \tag{96}$$

The affine state representation (68) implies

$$\begin{aligned}|x_t(\mathbf{m}_t)-x_t(\mathbf{m}_{t+1})|&=|\langle\boldsymbol{\phi}_t^x,\mathbf{m}_t-\mathbf{m}_{t+1}\rangle|\\&\le\|\boldsymbol{\phi}_t^x\|_\infty\|\mathbf{m}_{t+1}-\mathbf{m}_t\|_1.\end{aligned} \tag{97}$$

Combining the triangle inequality with (96) and (97) yields

$$\begin{aligned}d_{t+1}&\le\alpha d_t\\&\quad+(\alpha\|\boldsymbol{\phi}_t^x\|_\infty+\|\boldsymbol{\phi}_t^u\|_\infty)\,\|\mathbf{m}_{t+1}-\mathbf{m}_t\|_1.\end{aligned} \tag{98}$$

By (71),

$$\begin{aligned}\alpha\|\boldsymbol{\phi}_t^x\|_\infty+\|\boldsymbol{\phi}_t^u\|_\infty&\le\frac{\alpha^2}{1-\alpha}+\alpha\\&=\frac{\alpha}{1-\alpha}.\end{aligned} \tag{99}$$

Lemma 9 and $\|\mathbf{g}_t\|_\infty\le G_\alpha$ therefore reduce (98) to

$$d_{t+1}\le\alpha d_t+\frac{\alpha\eta G_\alpha}{1-\alpha}. \tag{100}$$

Unrolling from $d_1=0$ gives

$$\begin{aligned}d_t&\le\frac{\alpha\eta G_\alpha}{1-\alpha}\sum_{j=0}^{t-2}\alpha^j\\&\le\frac{\alpha\eta G_\alpha}{(1-\alpha)^2},\end{aligned} \tag{101}$$

with an empty sum when $t=1$. This proves (37).

Finally, $\widehat{u}_t=u_t(\mathbf{m}_t)$ and the feasibility of fixed SDAC policies imply

$$\begin{aligned}0\le u_t(\mathbf{m}_t)-u_t&=[u_t(\mathbf{m}_t)-\alpha x_t]_+\\&\le[\alpha x_t(\mathbf{m}_t)-\alpha x_t]_+\\&\le\alpha d_t.\end{aligned} \tag{102}$$

Combining (102) with (37) proves (38).

Both state–action pairs lie in $\mathcal{D}_\alpha$. Hence (5) and the two bounds just proved give

$$\begin{aligned}&c_t(x_t,u_t)-\ell_t(\mathbf{m}_t)\\&\quad\le L_x\big|x_t-x_t(\mathbf{m}_t)\big|+L_u\big|u_t-u_t(\mathbf{m}_t)\big|\\&\quad\le\frac{\alpha\eta G_\alpha L_x}{(1-\alpha)^2}+\frac{\alpha^2\eta G_\alpha L_u}{(1-\alpha)^2}\\&\quad=\frac{\alpha\eta G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}.\end{aligned} \tag{103}$$

Summing (103) over $t=1,\dots,T$ proves (39). $\square$

Proof of Theorem 1. Define the stability term

$$\mathrm{Stab}_T\triangleq\sum_{t=1}^{T}\left(c_t(x_t,u_t)-\ell_t(\mathbf{m}_t)\right).$$

Using $\ell_t(\mathbf{m})=c_t(x_t(\mathbf{m}),u_t(\mathbf{m}))$, the regret decomposes as

$$\mathrm{Reg}_T^{\mathrm{SDAC}}=\mathrm{Stab}_T+\sum_{t=1}^{T}\ell_t(\mathbf{m}_t)-\min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}). \tag{104}$$

We first bound the initial Bregman divergence. Fix $\mathbf{m}\in\mathcal{M}_H$. Since $\mathbf{m}_1=\frac{1}{H+1}\mathbf{1}$,

$$\begin{aligned}D_\Psi(\mathbf{m}\|\mathbf{m}_1)&=\sum_{i=1}^{H}m_i\log\big((H+1)m_i\big)\\&\quad-\|\mathbf{m}\|_1+\frac{H}{H+1}\\&=\sum_{i=1}^{H}m_i\log m_i+\|\mathbf{m}\|_1\log(H+1)\\&\quad-\|\mathbf{m}\|_1+\frac{H}{H+1}.\end{aligned} \tag{105}$$

Because $z\log z\le 0$ for $z\in[0,1]$,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1)\le\|\mathbf{m}\|_1\log(H+1)-\|\mathbf{m}\|_1+\frac{H}{H+1}. \tag{106}$$

The right-hand side is affine in $\|\mathbf{m}\|_1 \in [0,1]$, and its values at the endpoints are

$$\frac{H}{H+1} \le \log(H+1),$$
$$\log(H+1) - \frac{1}{H+1} \le \log(H+1).$$

Here the first inequality follows from $\log z \ge 1 - z^{-1}$ for $z \ge 1$. Therefore,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \le \log(H+1) \qquad \text{for every } \mathbf{m} \in \mathcal{M}_H. \quad (107)$$

Lemma 10, $\|\mathbf{g}_t\|_\infty \le G_\alpha$, and (107) imply, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned}\sum_{t=1}^{T}\big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) &\le \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T}\|\mathbf{g}_t\|_\infty^2 \\ &\le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}.\end{aligned} \quad (108)$$

Since (108) holds uniformly over $\mathbf{m}$,

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m}) \le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}. \quad (109)$$

The stability bound (39) gives

$$\mathrm{Stab}_T \le \frac{\alpha\eta T G_\alpha(L_x + \alpha L_u)}{(1-\alpha)^2}. \quad (110)$$

Substituting (109) and (110) into (104),

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{SDAC}} &\le \frac{\log(H+1)}{\eta} + \frac{\eta T G_\alpha^2}{2} \\ &\quad + \frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2} \\ &= \frac{\log(H+1)}{\eta} + \eta T\Gamma_\alpha.\end{aligned} \quad (111)$$

Differentiating with respect to η yields

$$-\frac{\log(H+1)}{\eta^2} + T\Gamma_\alpha.$$

The unique minimizer is therefore

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}. \quad (112)$$

At this value the two summands in (111) are equal, so

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T\log(H+1)}. \quad (113)$$

It remains to verify the dependence on α. As $\alpha \downarrow 0$,

$$\begin{aligned}\frac{G_\alpha}{\alpha} &= \frac{L_x}{1-\alpha} + L_u \longrightarrow L_x + L_u, \\ \frac{\Gamma_\alpha}{\alpha^2} &= \frac{1}{2}\left(\frac{G_\alpha}{\alpha}\right)^2 + \frac{G_\alpha}{\alpha}\frac{L_x+\alpha L_u}{(1-\alpha)^2} \\ &\longrightarrow \frac{(L_x+L_u)^2}{2} + L_x(L_x+L_u) > 0.\end{aligned}$$

Therefore $\sqrt{\Gamma_\alpha} = \Theta(\alpha)$ as $\alpha \downarrow 0$. As $\alpha \uparrow 1$, write $\varepsilon = 1-\alpha$. Then

$$\begin{aligned}\varepsilon G_\alpha &= \alpha(L_x + \varepsilon L_u) \longrightarrow L_x, \\ \varepsilon^3\Gamma_\alpha &= \frac{\varepsilon^3 G_\alpha^2}{2} + \alpha(\varepsilon G_\alpha)(L_x + \alpha L_u) \\ &\longrightarrow L_x(L_x+L_u) > 0.\end{aligned}$$

Thus $\sqrt{\Gamma_\alpha} = \Theta((1-\alpha)^{-3/2})$ as $\alpha \uparrow 1$. □

Proof of Corollary 2. Fix $\mathbf{q} \in \mathcal{Q}_\alpha^\downarrow$ and use the SDAC comparator $\mathbf{m}_{H_T}(\mathbf{q})$ from Lemma 4. Theorem 1 and (24) give

$$\begin{aligned}J_T^{\mathrm{on}} - J_T(\mathbf{q}) &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_{H_T}(\mathbf{q})T \\ &\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.\end{aligned}$$

The bound is uniform in $\mathbf{q}$, so taking the infimum comparator cost proves (40). Finally, $H_T \ge \log T/\log(1/\alpha)$ implies $\alpha^{H_T} \le 1/T$, while $H_T = \Theta(\log T)$ implies $\log(H_T+1) = O(\log\log T)$. □

Proof of Corollary 3. For fixed k, Theorem 1 applied to $\mathbf{m}_{H_T}^{\mathrm{prop}}(k)$, followed by (26), gives (41). Corollary 1 shows that the geometric allocation sequence $\mathbf{q}(k)$ belongs to $\mathcal{Q}_\alpha^\downarrow$, so Lemma 4 gives $\mathrm{Tail}_{H_T}(\mathbf{q}(k)) \le \alpha^{H_T+1}$. This makes the bound uniform in k; minimizing over k proves the stated finite-time and asymptotic guarantees. □

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